Title and Source
Introduction of ER Model
Source: GeeksforGeeks

Main Idea
The post explains the Entity-Relationship (ER) Model—a conceptual approach to database design. ER diagrams visually represent entities, their attributes, and relationships, helping map real-world data structures before implementation.

Key Details

Entities and Entity Sets
Entities are real-world objects (e.g., Student, Course) represented by rectangles. An entity set is a group of similar entities.

Types of Entities

- *Strong Entity:* Has a primary key; independent.
- *Weak Entity:* Lacks a unique identifier; depends on a strong entity.

Attributes
Attributes describe entities and are shown as ellipses.

- *Key Attribute:* Uniquely identifies entities (underlined).
- *Composite:* Made of sub-attributes (e.g., Address = Street + City).
- *Multivalued:* Can hold multiple values (e.g., Phone numbers).
- *Derived:* Calculated from other data (e.g., Age from DOB).

Relationships
Relationships link entities and are shown as diamonds. A relationship set is a group of such links.

Relationship Degree

- *Unary:* One entity (e.g., Person marries Person)

- *Binary:* Two entities (e.g., Student enrolls in Course)
- *Ternary/N-ary:* Three or more entities

Cardinality

- *One-to-One:* One entity maps to one
- *One-to-Many:* One maps to many
- *Many-to-One:* Many map to one
- *Many-to-Many:* Many map to many

Participation

- *Total:* Every entity must participate (double line)
- *Partial:* Optional participation (single line)

Symbols Used
Rectangles for entities, ellipses for attributes, diamonds for relationships, double ellipses for multivalued attributes, and double rectangles for weak entities.

Steps to Draw ER Diagrams
Identify entities, define relationships, add attributes, assign keys, remove redundancies, and review for clarity.

Relevance
ER diagrams are essential for structuring the database system of GradeGator enabling clear mapping of users, courses, submissions, and grades.

Reflection
The post clarified how logical design underpins efficient database systems. Understanding entity types, attributes, and cardinality will directly inform how we design data models. It also raised interest in managing more complex (ternary) relationships.